PROMOTING EMPATHY IN DECISION-MAKING BY TURNING AGENT-BASED MODELS INTO STORIES USING LARGE LANGUAGE MODELS

Anonymized

ABSTRACT

While Agent-Based Models (ABMs) support decision-making processes by simulating 'what-if' scenarios, their outputs may not engage decision-makers in thinking of potential intervention from the viewpoint of the people impacted by the decision. Given the importance of empathy in decision-making, there is a gap between simulation outputs and the human element in policy formulation. By combining Large-Language Models (LLMs) and domain-specific knowledge, we transform simulation data into narratives that provide rich descriptions of an agent's simulated journey. We perform automatic and manual evaluations with GPT-4 on three previously developed ABMs, along with different prompts and parameter settings. Across all cases, generated narratives have a high readability level, adequately incorporate human elements into the story, and faithfully follow an agent's simulated dynamics. Integrating such text into simulation environments would be a valuable future step, which requires research on style (e.g., to avoid a fantastical tone) and a story's relevance to decision-making activities.

Keywords: agent-based model, decision-making, GPT-4, natural language generation.

1 INTRODUCTION

Agent-Based Modeling are routinely used to investigate complex social decisions such as who should be hired as a faculty [1], whether a person shall receive governmental assistance such as food stamps [2], or when to execute individuals involved in terrorism [3]. Decision-support systems can make it easier to manage a public benefit system or faculty hiring since they shield decision-makers from personally facing the people whom they may reject or hurt. When using an ABM, decision-makers can simulate the effects of a decision and use statistical analyses to examine simulation outputs. Such analyses and spreadsheets of simulation outputs widen the disconnect, as the individuals who make critical decisions may not even be able to *feel* as if they were among the people impacted by the decision. Without thinking of how individuals can be impacted, decision-makers can lack *empathy*, that is, "the capacity to form moral evaluations from another person's perspective" [4]. The importance of empathy has been stressed when making decisions about interconnected systems, as empathy fosters cooperation [4].

Given *(i)* the importance of empathy in decision-making processes involving human lives and *(ii)* that spreadsheets or analyses of simulated agents do very little to foster empathy, we propose to bridge this gap by automatically transforming the quantitative data characterizing an agent into a relatable *story*. For example, consider a common type of ABM in which we manage the evacuation of a population [5, 6]. We could look at the effect of an intervention by monitoring the number of people who are safe over time, or by inspecting simulation traces to see that agents are successfully evacuated one after another. We may also read that one agent, Sarah, a mother of three, had to evacuate without a car and experienced high stress levels. This adds a new dimension to the effect of the evacuation protocol and it could make us nuance its overall 'efficiency'. Such *humanizing techniques* are commonplace in journalism [7], where articles routinely start by telling us about a specific individual before pivoting onto community or societal matters.

The technical contribution in our work is to *automatically generate stories for agents*. By blending domain expertise and Natural Language Generation via Large-Language Models (LLMs), we assist in humanizing simulation outputs by generating narratives that capture the essence of the agents' experience. Although there is a growing interest in using LLMs together with simulations [8, 9, 10], our study is the first to narrate the agents' simulated journeys via LLMs. Our work complements the recent study of Aoki *et al.*, in which the authors also used agents to create narratives, but the goal was to generate stories (i.e., creative writing) [11]; hence, their *agents only served to support the creation of narratives* whereas our narratives seek to ultimately support decision-makers in appreciating an agent's simulated journey. We demonstrate our approach by applying GPT-4 to three published ABMs: evacuation during a fire or flood [5], hurricane evacuation [6], and human migration [12]. To support replicability efforts in simulation and provide full transparency into our study, all scripts and data involved in our demonstration are openly available on a third-party anonymized repository at https://doi.org/10.5281/zenodo.10552705.

The remainder of this paper is structured as follows. We assume that readers are familiar with notions of agent-based modeling but may not have expertise in text processing. Thus, section 2 provides background information on the relation between text and simulations, as well as the use of LLMs to promote empathy. Then, section 3 summarizes the key features of the three ABMs used as case studies and details the generation of agent stories by using GPT-4 as LLM. We present our results in section 4.4, using both automatic metrics (e.g., readability) and human evaluations (e.g., does the text reflect the agents' dynamics?) to assess the quality of our proposed system depending on the ABM, desired length of a story, and temperature setting (i.e., the 'creativity' parameter for GPT-4). Section 5 briefly contextualizes the importance of our results and the limitation of our current approach. Examples of stories are provided at the end of the article.

2 BACKGROUND

2.1 Agent-Based Models and Natural Language Processing/Generation

While this paper is the first one to generate *text from ABMs*, there are several studies that blend ABMs and narratives [13] or focus on creating *ABMs from text*. Although our goal is not to turn an ABM 'back' into the descriptions used in these studies, they are nonetheless valuable to assess how an agent can be described. In the work of Padilla and colleagues [14, 15], an ABM is divided into three components: the acting agent who performs actions, the attributes or 'contextual factors' of the agents, and the rules or mechanisms that govern the actions. The systems created by the authors takes as input a narrative provided by a domain expert, dissects its content (e.g., nouns, adverbs, verbs), and produces a conceptual model to highlight actors, attributes, and mechanisms. The authors note that the creation of a conceptual model from text is imperfect, due to e.g. inconsistencies in natural language and differences in writing styles. We emphasize that the imperfections are intrinsic to the task rather than a reflection of the particular algorithm employed. For example, there are also notable discrepancies when participants are tasked with producing conceptual models, given the same textual evidence [15, 16]. Participants may interpret the narrative differently or consider additional aspects that were not explicitly provided in the text. This suggests that any tool bridging simulations and narratives may require some human intervention, hence it can be useful to provide parameters that allow modelers to adapt the tools to their specific needs.

Given the advent of LLMs across simulation areas, several works have considered and empirically evaluated the use of LLMs to generate or improve ABM implementations. Results have been mixed, which is expected in a nascent field that does not yet have benchmarks and employs very different approaches to assessment. In one study, researchers noted that the process of generating executable model from a narrative may not be fully automated *at present* [17]. This limitation may partly stem from a lack of familiarity of an LLM with simulation languages, which can be addressed as more data is collected. In another study, the authors concluded that their LLM-based framework *improved* model quality [18]. While these results may seem

at odds with one another, they considered different languages (`NetLogo` vs. `Python`) and had different use cases. Beyond the matter of gaining the ability to generate production code with LLMs, there is an important matter in integrating these new forms of automation into the existing simulation workflow. A study that created a smart conversational agent (via LLMs) to assist with modeling in `NetLogo` noted that experts were more inclined than novices to use LLMs in their modeling workflow [19].

A separate strand of research in 'Generative Agent-Based Modeling' is combining Generative AI (i.e., LLMs) with ABMs to create a hybrid model. The rationale for this integration is to improve communication expressiveness (instead of using agent communication protocols) and support more autonomous, dynamic decision-making processes [20]. Dedicated libraries such as `Concordia` facilitate this new approach [21] and interested readers may further explore this area through a dedicated tutorial by Ghaffarzadegan *et al.* [8]. An extensive example was recently provided in which the agents represented hypothetical Alien Civilizations who interacted via their LLMs to learn about each other and eventually coexist [22]. In another instance focused on misinformation, agents had roles such as 'Russian Troll' or 'European leader' and the LLMs gave them the ability to reflect on information, potentially debunking fake claims [23]. The area of game development has been noticeably prolific in creating new products that include LLMs in some aspect of an agent. A survey from April 2024 helps to situate these recent advances by breaking down an agent's decision-making module into modules (e.g., memory, planning, perception) and examining the use of LLMs for each module (e.g., episodic memory, semantic memory, procedural memory) [24].

2.2 Empathy and Large Language Models

Empathy builds on two components: *affective empathy* gives the capacity for emotional responses when interacting with others, while *cognitive empathy* allows to understand the experiences of another person [25]. In this paper, we use NLG to promote cognitive empathy in decision-making. Cognitive empathy has been abundantly studied in decision-making. For example, patient-centered care requires empathy from medical practitioners when they interact with patients, either directly or indirectly (e.g., via narratives of a patient's case). In a recent study, medical students were provided narratives of patients diagnosed with schizophrenia. The study showed that students were able to demonstrate cognitive empathy towards the patient, based on the narratives [25]. In another report, a design firm working on a cancer care center reported that written storytelling (along with other narrative techniques) helped participating students in generating design solutions that *support the whole person* instead of only a few quantitatively measured facets of the patient's experience [26]. Since stories promote empathy, can Large Language Models (LLMs) craft such stories?

A systematic review of seven studies using LLMs for empathy concluded that LLMs can exhibit cognitive empathy when responding [27]. These findings are reinforced by the study of Huang *et al.* [28], who examined five LLMs in 428 situations where an LLM expressed its agreement with a set of statements on a scale from 1 to 5. The same situations were presented via crowdsourcing to more than 1,266 human evaluators to compare emotional responses between LLMs and people. The authors observed that "LLMs can generally evoke appropriate emotions in response to specific situations". At the same time, they reported that LLMs generally show more intense emotions than people, with a few notable exceptions such as a lack of jealousy/envy in LLMs compared to people (e.g., humans are annoyed when realizing that a peer bought the same item for cheaper – LLMs don't mind). Of particular importance for our own study, we note that the benchmark created by Huang *et al.* relied on *situational prompts*, where scenarios were provided to an LLM so that it 'imagined' being in a situation. These situations were extracted from the literature: they were *not* automatically connected to any model, such as an agent-based simulation.

3 METHODS

Our process is summarized in Figure 1. We start by selecting three ABMs from `CoMSES.net` with the criteria that they must be *(i)* associated with a peer-reviewed publication (as a form of certified model documentation) and *(ii)* depict a human system. The three ABMs are summarized in subsection 3.1. If the agents already have explicit features that could tell their stories, we proceed to simulating the model. Otherwise, we add a feature to track each agent's journey. Simulation outputs consist of a large spreadsheet which tracks the agents' values over time tick. Through a pre-processing stage, we normalize and summarize the data in linguistic terms for the sake of crafting a journey instead of narrating a table. Simulation and pre-processing are detailed in subsection 3.2. Finally, we pass a prompt with the agent's summary to GPT-4 (subsection 3.3, which produces a narrative. Narratives are evaluated in the next section, which details our design of experiments and evaluation metrics.

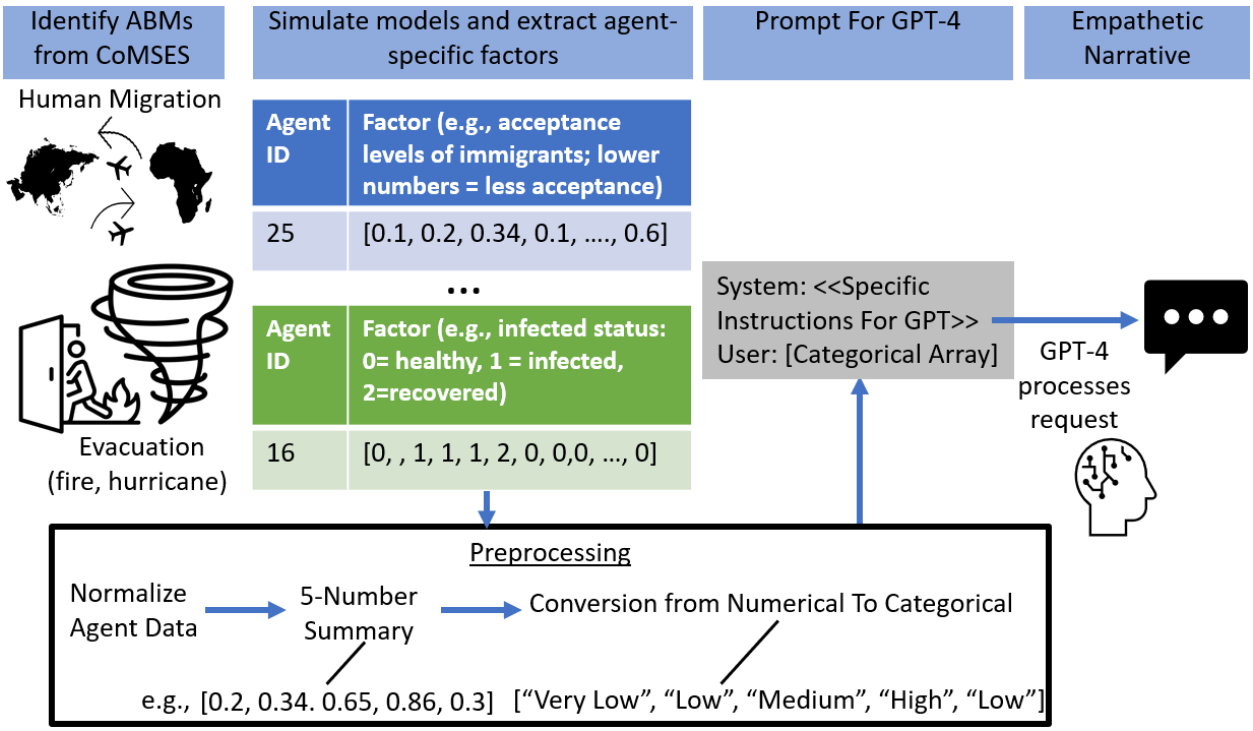

Figure 1: Key steps of our process to generate narratives from models.

3.1 Three Peer-Reviewed Open-Source ABMs

MigrAgent (Figure 2) simulates an agent's decision to emigrate from their host country to a different one, based on various factors, such as skills, economic status, ethnicity, and interactions with the local population [12]. The model was developed to study the causes of a phenomenon known as 'brain drain,' where a country's highly skilled workers emigrate. Locals and migrants interact with a probability linked to their conservatism. If these interactions result in acceptance, the agent's conservatism decreases. Conversely, if the interaction results in rejection, the agent's conservatism increases. The agents in the model have a factor called *acceptanceLevel*, which computes the overall acceptance of locals and immigrants with respect to each other over time. Given that migration can be a challenging step for individuals seeking to make a living and support their families, supporting empathy in decision making can result in making decisions that consider the migrants' journeys more holistically.

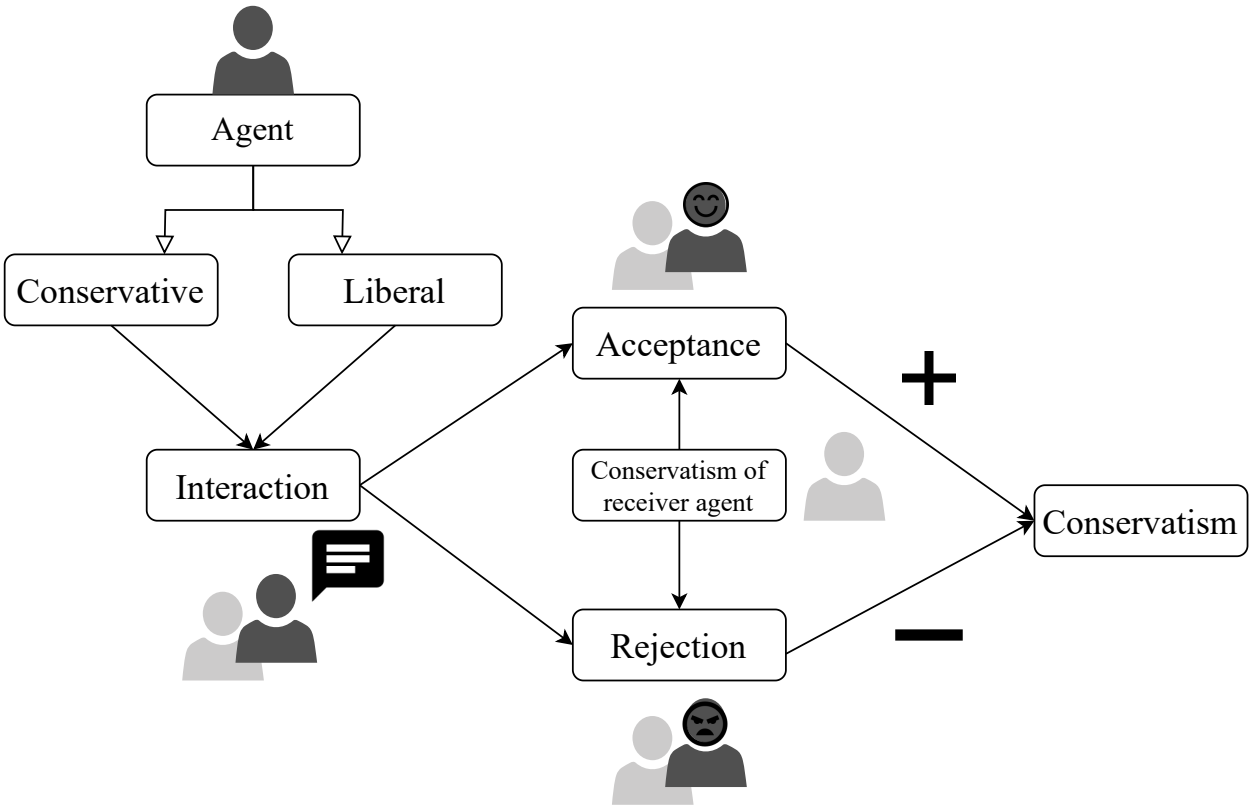

Figure 2: Simplified flowchart of agent actions for **migragent** [12].

CHIME (Figure 4) illustrates the movement of citizens in Florida in response to a hurricane threat, aiming to simulate the decision-making process people undergo to protect themselves during a storm [6]. An agent's movements depend on various traits, including their surrounding environmental cues, disabilities, connections to a broadcaster, and proximity to the storm. CHIME has a risk-estimate factor for each agents, which depends on the 'perception' of an agent as it potentially gets influenced by interacting with neighboring agents and/or broadcasting agents. The higher risk estimate leads to higher risk perception, and thus higher possibility that the agent will evacuate. This model could help policymakers in determining locations for broadcasting weather updates, allocating emergency personnel as needed, and issuing evacuation orders in specific areas. Therefore, fostering empathy for those who can be affected by the storm is crucial.

BNE (Figure 4) aims to study optimum approaches that a collective group of pedestrians may adopt when encountering perilous situation. The model contains agents (i.e., pedestrians) and two exits, located on the right edge and left edges of the simulation grid. The agents act as barriers to each other as they try to approach the exit and evacuate from the danger (e.g., fire, flood) [5]. Understanding macroscopic emergence of behaviors of agents as a collection is dependent on their micro/individual goals. Therefore, studying these emergence patterns can assist with potential intervention (e.g., sending fire brigades during fire evacuation) strategies. The behavior of agents is determined as the user selects one of three available sub-models: paths with the minimum number of obstacles, following a local leader agent (designated at random), or using a utility function computed for each agent based on distance and comfort. Agents will choose a path depending on congestion level, distance to the exist, and other agent movement in their Moore neighborhood. The model is purely mathematical as agents make decisions based on the principles of bounded rationality or achieving Nash Equilibrium, depending on which of the three models is chosen. However, humans could make irrational decisions in these stressful situation (e.g., being paralyzed from fear). Hence, there is a need

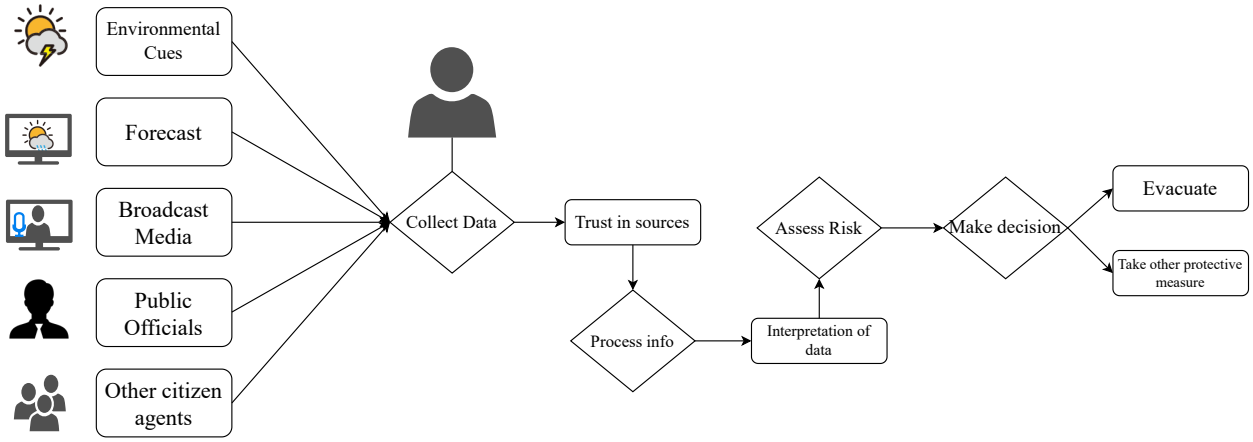

Figure 3: Simplified flowchart of agent actions for **CHIME** [6].

to connect the mathematical output generated from this model into an empathetic narrative that can capture the scenario in the perspective of the agent.

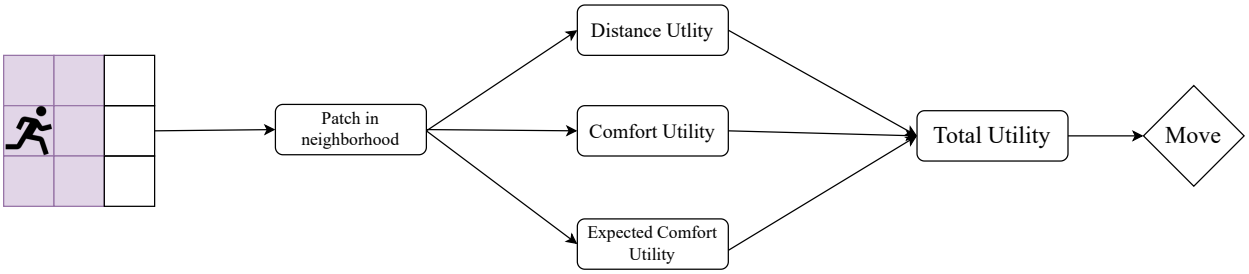

Figure 4: Simplified flowchart of agent actions for **BNE** [5].

3.2 Simulation and Prompt Pre-Processing

To transform an agent's journey into text, we first need to *track* this journey through attributes in the model. For **MigrAgent**, we use the existing *acceptance levels* that computes how the locals accepted the incoming migrants over time. For the **CHIME** hurricane model, we use the existing *riskPerception*, which is the calculated risk of the incoming hurricane as perceived by the agent from its interactions with other agents. The **BNE** model focused on the decision-making processes of the agents, but it did not track their experiences. We thus use this case to illlustrate that a modeler can make a variable explicit, in order to link an ABM to the generation of empathetic stories. We added a numerical feature *stressLevel* to the agents, which is updated during the simulation as shown in Table 1.

We run **MigrAgent** for 800 ticks to allow for enough time for the locals and migrants to "mingle", such that the model accounts for adjustments that would be expected for a different ethnicity, tradition, or culture. We run **CHIME** for 138 ticks as that is the overall time span for the hurricane to pass over the region. We run the **BNE** evacuation model for 50 ticks, so that agents have a tight yet feasible amount of time to evacuate a building. After each simulation is run, we have an array of values for each agent. For example, with BNE, we have a dictionary with each agent as key and their stress levels over time as associated values.

Although GPT-4 has a stronger ability at handling scientific data than its successors, pre-processing is still useful. When modelers prepare a query *based on their understanding of the context*, it decreases the risk of errors from GPT-4. We thus pre-process the array by using three techniques. First, we *normalize* the

Table 1: Stress Levels of agents. The level is set to 0 if an agent manages to evacuate.

Ticks	Stress Level	Reasoning
0 (inclusive) to 10 (exclusive)	2	Agents start with an initial stress as they are trying to evacuate in a stressful setting
10 (exclusive) to 20 (inclusive)	5	If agents cannot leave the area between 10 and 20 ticks, then they become more stressed.
20 (exclusive) to 30 (inclusive)	10	If agents spend even more time without evacuating, then they get more stressed.

data, as done in prior work where price data was normalized before asking an LLM to perform financial forecasting [29]. Normalization consists of scaling the values of the factor of interest between 0 and 1. Second, in line with prior works on streaming data, we focus on *aggregate statistics* instead of providing every data point [30]. Sending less data to GPT will lower the cost of the queries and it also helps to abstract stories by focusing on trends, instead of narrating what happened to an agent at every time tick. We reduced our array sizes to 5 elements for all three models. The elements of the array were the categorical bins representing the agent's qualitative factor levels. The first element of the array was at the first time tick. The next three elements correspond to the 25th, 50th, and 75th percentiles across time ticks (i.e., lower quartile, median, upper quartile). The final element is the value on the last simulation tick. Third, GPT handles textual information better than numeric information. We converted the content of the numerical array into *linguistic variables* based on the following bins:

- "Very Low" : Values between 0.0 and 0.2
- "Low" : Values between 0.2 and 0.4
- "Medium" : Values between 0.4 and 0.6
- "High" : Values between 0.6 and 0.8
- "Very High": Values between 0.8 and 1.0

An example of the original array, and the intermediate normalized array, and the categorical array for BNE model (Figure 5) indicates the transformation for the final array used as part of the prompt for GPT.

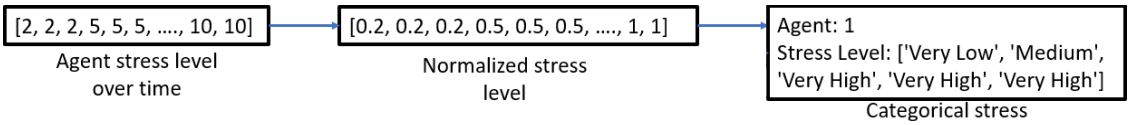

Figure 5: Example of array transformation.

3.3 Prompt engineering

We wrote one prompt for each of the three ABMs, as shown below. The design of a prompt can include several elements such as context, style, (counter-)examples, or procedural guidance (e.g., eliciting step-by-step reasoning by chain-of-thought prompting). There is growing evidence that using more prompt engineering techniques does not necessarily produce better answers [31, 32], hence we focus on context and style. Our prompts provide *context* on the ABM, for example by conveying that it is about acceptance of immigration or emotions related to evacuating. Generative AI can be understood as a *style engine*, which learns certain encoding styles from its training data and can transfer them [33]. Prompts can thus include keywords to deliberately evoke particular styles. Keywords are a delicate balance: the notion of a style can be fuzzy and overly generic (e.g., what does it mean to write 'like a journalist'?), and LLMs are easily

prone to excess that will only be noticed by the modelers when reading the response. When asking GPT-4 for empathy, we observed results that were melodramatic. For example, the model of evacuation produced sentences such as "the fearful beats of stress amplified to a deafening pitch, a relentless echo bouncing off the stone-cold walls of the tunnel" or "the stress began to spiral, like the needle on an old vinyl, repeatedly playing the same beats of worry". As our intent is not to generate horror fiction that will instill terror in the hearts of decision-makers, we instructed GPT to provide empathy *while minimizing dramatization*. To make the agents more relatable, we gave them *a name*, by asking GPT to either 'Choose a male name' (for MigrAgent and BNE) or 'Choose a female name'.

The systematic review on empathy and LLMs by Sorin *et al.* noted on several occasions that answers created by GPT were *too lengthy* [27]. This may incorrectly be perceived as empathetic, as the accumulation of details afforded through a longer story can make a character seem more alive. We thus create three versions of each prompt by adding a final instruction, which is "Provide your response in one concise paragraph.", "Provide your response in three concise paragraphs.", or left blank (no restriction). We emphasize that our goal is not to transform a *single* agent into a complete and lengthy report, but rather to provide a sufficient vignette of the agent to promote empathy. An Agent-Based Model consists of several agents, hence users would examine several narratives to appreciate the journey of the agents.

MigrAgent Prompt

As an agent in an Agent-Based Model (ABM) simulating immigrant interactions in a diverse society, you assume the role of a local named «`AGENT NAME`». In this model, you represent the majority of natives who hold liberalist notions. Your role is to embody empathy and understanding towards immigrants from a different country, reflecting the sentiments of openness and acceptance that are prevalent among your fellow locals. Craft a narrative that conveys your character's empathetic outlook, highlighting the importance of harmonious coexistence without using numbers and keeping the dramatization to a minimum. The array that will be provided to you will contain an agent's acceptance levels over time. «`TEXT LENGTH`» «`ARRAY OF LOCAL ACCEPTANCE OF AGENT`»

CHIME Prompt

Imagine you're in the shoes of an agent living in Florida, embodying the role of a woman named, «`AGENT NAME`». Your task is to simulate her perception of risk as she receives information from neighbors and news broadcasts about an impending hurricane, code-named CHARLEY. The accuracy of her risk perception is crucial, as it influences her decision to evacuate, especially since the hurricane is expected to pass very close to her location. Craft an empathetic narrative that delves into her emotional journey, emphasizing her response to the evolving risk. Ensure you avoid using numerical data and maintain a narrative tone that minimizes dramatization while capturing the essence of her decision-making process. The array that will be provided to you will contain her perceived risk assessment level in regards to her safety over time. «`TEXT LENGTH`» «`ARRAY OF RISK PERCEPTION OF AGENT`»

BNE Prompt

Imagine you are immersed in an Agent-Based Model (ABM) simulation, focusing on pedestrian evacuation from a tunnel equipped with two exits. In this scenario, you step into the shoes of a character named «AGENT NAME», who is an agent in the simulation. Your role is to grasp the intricate dynamics of stress levels within this agent as he navigates the evacuation process. Craft an empathetic narrative that unveils his emotional journey, capturing the nuances of his experience without resorting to numerical data and maintaining a narrative tone that minimizes dramatization. The array that will be provided to you will contain his stress levels over time. «TEXT LENGTH» «ARRAY OF STRESS LEVEL OF AGENT»

4 EXPERIMENTAL EVALUATION

4.1 Goal and Design of experiments

Our objective is to assess the 'quality' of the answers produced by GPT. A prior review found that answers needed a high reading level, which limited their applicability [27]. We thus use automatic quantitative measures of readability, known as the Flesch scores. We also perform a manual assessment by reading the text, to evaluate aspects such as the faithfulness and fluency of the writing. These scores depend on the case study, the temperature setting of GPT (which introduces randomness), and the desired length of the output. We thus created a full factorial design of experiments consisting of all combinations of: 3 agent-base models, 5 temperature settings (0.00, 0.25, 0.50, 0.75, 1.0), and 3 text lengths (no restriction, 3 paragraphs, 1 paragraph). Experiments are repeated four times due to randomness. This yields of total of $3 \times 5 \times 3 \times 4 = 180$ texts.

4.2 Programming Environment, Libraries, and Model Versions

The three ABMs are **MigrAgent** version 1.4.0, **CHIME** version 2.1.0 for hurricane evacuation, and **BNE** version 1.0.0 for pedestrian evacuation. We use NetLogo (version 6.1.1) to run simulations for these three models. Our code is written in Python to connect with NetLogo, process the simulation outputs, and generate contextual prompts to send the simulation data to GPT (version gpt-4-1106-preview) for generation of empathetic narratives. All work is conducted within anaconda (version 23.9.0) virtual environment to avoid errors generated from package incompatibilities. The core packages are OpenAI 0.28.1 (for GPT), numpy 1.26 (to represent the agent's array of features), textstat 0.7.3 (to automatically measure the readability of text), pynetlogo 0.5.1 (to run the NetLogo model from Python), openpyxl (to access the spreadsheet of results from NetLogo), and seaborn 0.12.2 (to visualize results). Our script is provided as a Jupyter Notebook as part of our repository at https://doi.org/10.5281/zenodo.10552705.

4.3 Automatic and Manual Measures of Quality

We **automatically** evaluate the readability of the text generated by GPT-4 by using two Flesch Scores. The *Flesch Readability Score* ranges from 1 to 100 and assesses the grammatical structure to assign a score to the provided text. A high score indicates high readability. The *Flesch-Kincaid Score* estimates the school education level required to be able to understand the text (e.g a score of 5 means that a fifth grader can understand it). We also manually evaluated the 180 prompt responses, by attributing points as follows:

1. Is there a human (name and human traits) (1 point)? In the prompts, we specified the gender we assigned to an agent, and hence evaluated for correctness if the name given to the agent was correct with respect to the gender.
2. Does the output reflect the initial, final, and trends of the agent's dynamics per the array? (2 points) This aspect deserves more points because it reflects the ability of GPT-4 to understand the given data and produce an empathetic narrative corresponding exactly to the data.
3. How many instances of poor writing? (1 point) As in prior works [34, 35], we deduct $\frac{1}{\text{number of sentences}}$ per issue. For example, if one of four sentences exhibits poor fluency, we deduct 0.25.

The manual assessment was performed by two human evaluators, consisting of the first two authors. For consistency in evaluation, the evaluators first independently evaluated a sample of 20 responses of the 180 total responses and discussed their evaluations to arrive at a consensus. The rest of the 160 remaining responses were then divided equally, with each evaluator evaluating 80 responses. To support further independent analyses of our results, we have made them publicly available in our permanent repository.

4.4 Results

We generated heatmaps to evaluate the performance of GPT for each of the ABMs and the parameters of interest. We used averaged manual evaluation scores produced by our human evaluators and Flesch Reading Ease score (Figure 6). Our detailed results at the level of each prompt are available on https://doi.org/10.5281/zenodo.10552705. For all ABMs there is an optimal combination of temperature and text length that produces the highest performance for Flesch Reading Ease scores. This optimal combination is achieved at a temperature of 0.5 and in the absence of text restriction. Intuitively, it means that GPT does *better* when given some freedom to deviate from its training data and no constraints on length. Under the optimal combination, the reading ease either qualifies as 'plain English' (scores 60–70; 8th and 9th grade) or 'fairly easy' (scores 70–80; 7th grade). The average manual scores are generally high, indicating that the narratives generated by GPT were well written and faithfully reflected the dynamics of an agent. Restricting the narrative to a single paragraph has a negative impact on manual scores and on readability, which suggests that a limit of three paragraphs is a reasonable trade-off.

Our detailed manual scores show that the output almost always (97.22% of cases) presented anthropomorphic attributes as requested (i.e., name and human traits). The output tended to accurately reflect the initial, final, and trends of the agent's dynamics (91.67%). The quality of writing was more heterogeneous, as only half (50.55%) of the instances had a fluency score higher than 0.8 on a scale of 0 to 1. A frequent error was the verbatim use of linguistic qualifiers, as exemplified here: "However, the ebb and flow of cultural integration posed challenges, and I found my acceptance waning, reflected in a period where my openness was 'Low'. Yet within me persisted a will to understand and grow, and over time, I rediscovered the strength in our communal threads, lifting my acceptance back to 'High'." In contrast, highly rated outputs were able to weave the agent' states into a story: "The shared experiences with immigrant friends, learning about their customs, and collaborating on community projects heightened the level of acceptance and admiration I felt for the resilience and contributions of these individuals. From supporting immigrant-owned businesses to celebrating various cultural festivities, the barriers slowly dissolved, creating bonds stronger than I had anticipated." Additional samples are provided in tables 2 and 3, where undesirable outputs are shown in red.

5 DISCUSSION

Given the evidence that empathy is important in decision-making, we demonstrated the feasibility of generating varied narratives of an agent's journey given its simulation output and a brief description of the model's

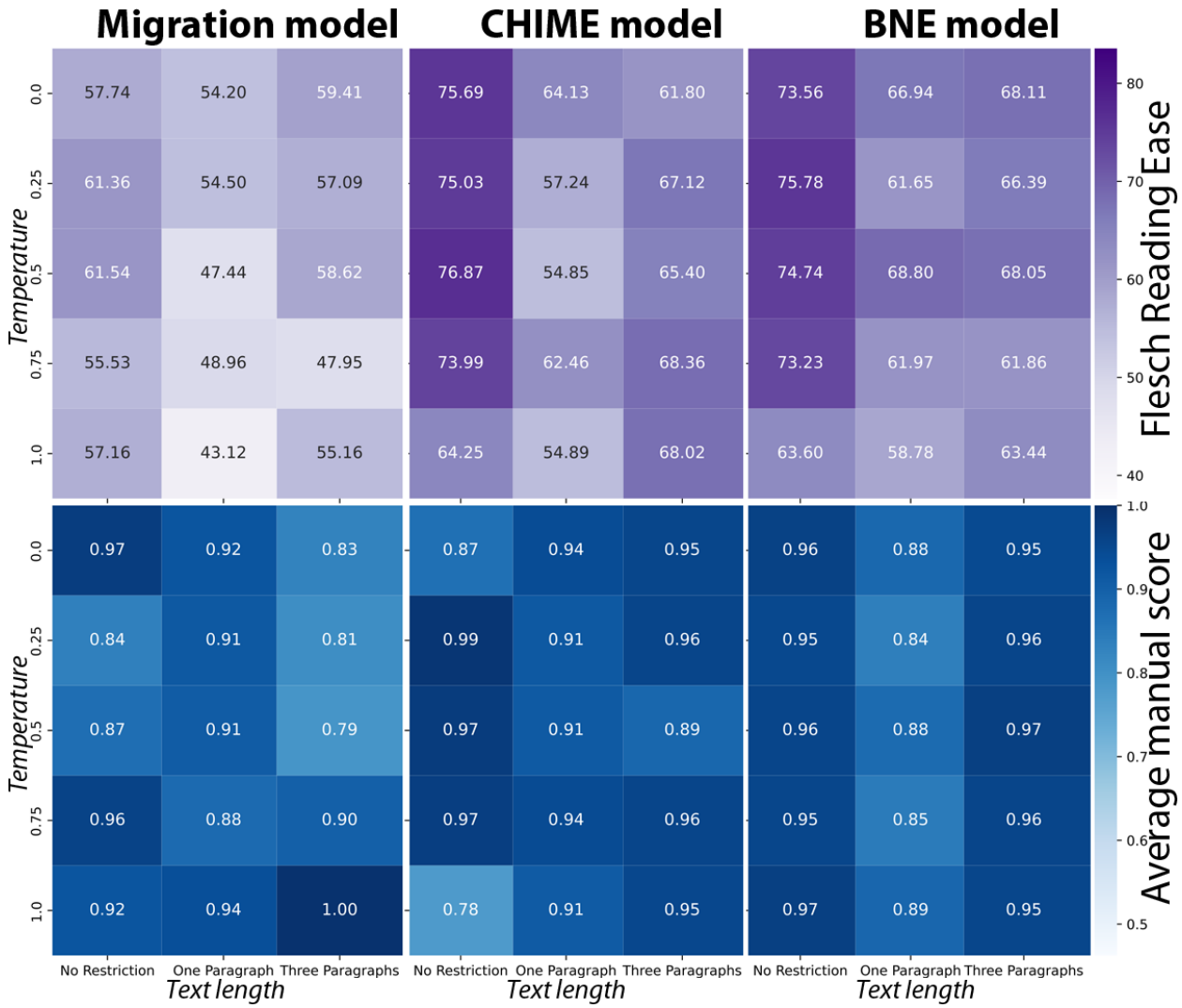

Figure 6: GPT performance for human and Flesch scores (higher values mean better performance).

context. Once suitably optimized[1], the outputs have good readability levels, display desirable anthropomorphic features that can help a reader connect with the story, and they generally follow the dynamics of an agent over the simulation. Although the narratives produced by GPT-4 provide a vivid description of an agent's actions, we note the flowery literary style of the writing on several occasions. For example, in the context of an ABM for migration, the agent observed that "neighborhood festivals were kaleidoscopes of color, and language was no barrier to the friendships that blossomed like spring flowers". This form of writing may promotes a fantastical tone that detracts from the importance and realism of the modeled scenario. In future works, it would be of value to evaluate additional prompts that guide GPT-4 to use simpler language. This may be achieved by including terms such as 'concise', 'modern', or 'normal' to describe the tone, as well as 'explicit' or 'direct' to describe the language. Further research may also evaluate different level of details. At present, our approach can distinguish values across agents: the narrative of an agent with a final value of

[1]Our work shows how to optimize the narratives on three different models. By providing these case studies along with our code, we hope to lower the level of expertise in prompt engineering that would be required from other modelers. However, modelers would still have to identify the appropriate attributes in a model that would be used for storytelling, which requires familiarity with the application domain and the model's design. Such attributes may not be available in all types of models.

Table 2: GPT Results. We found no instances of GPT producing wrong information. We highlighted instances where the subject perceives *their own* risk level as high or low because humans neither talk about risk in 'levels' nor do they usually express themselves in a style such as 'my risk perception is high'.

Model, Input, and Parameters	GPT response
Model: **MigrAgent** Input: Local Acceptance: ['Very High', 'Very High', 'Very Low', 'Medium', 'Medium']	As Lucas, I've always believed our community's strength lies in its diversity. Initially, my neighbors and I welcomed the newcomers with open hearts, eager to share our traditions and learn from theirs. However, when a wave of misunderstanding swept through us, driven by a few isolated incidents, my acceptance wavered, casting a shadow of doubt over my convictions. But with time and dialogue, I found a middle ground, recognizing that fear stemmed from the unknown. Now, as I strive to mend these rifts, I remain steadfast in my commitment to empathy, working tirelessly to ensure that we do not just coexist, but thrive together, celebrating the tapestry of cultures that enriches us all.
Model: **CHIME** Input: Risk Perception: ['Very Low','High', 'Medium', 'Medium', 'Low']	As Sophia peered out from her porch at the Florida sky, a gentle breeze tousled her hair, betraying no hint of the turmoil that was Charley, the hurricane churning towards her home. Initially, she felt a calm assurance; hurricanes had come and gone before, leaving her world untouched, her risk perception very low. But the chatter of concerned neighbors, coupled with the urgent tone of news broadcasts, soon seeded worry in her heart, elevating her sense of danger to high. She considered the cherished memories woven into the very fabric of her home, and her resolve wavered, the risk now a tangible medium in her mind, not once but twice, as she weighed the value of staying against the relentless approach of the storm. Yet, as Charley's path became clearer and her community's preparations more robust, Sophia's fear ebbed to a low, replaced by a cautious optimism. Trusting in the solidarity of her neighbors and the sturdiness of her home, she chose to weather the storm, her perception of risk a guidepost to the decision to stand firm in the face of Charley's wrath.

0.5 will differ from an agent with a final value of 0.12. However, some agents hold a low value for longer, or oscillate between states. Capturing such dynamics will need to strike a trade-off between the complexity of the text and its usefulness to guide decision-makers. Future evaluations could also include a comparison of generated narratives with traditional spreadsheets, as well as an analysis of decisions to identify how and why the empathetic presentation of agents acting in a rule-based manner matter could make a difference.

There is evidence that narratives can have an impact on decision-making, but it is not necessarily a *good* impact. A systematic review found that stories could sway public opinion [36], which echoes the importance of using 'anecdotes' in policy making [37]. However, the same review also reported cases where excesses in a narrative had counter-productive effects [36]. An experimental study with decision-makers would be of particular interest, by comparing their decisions based on either (i) a model's interface or (ii) a sample of stories. The integration of stories with simulation interfaces is a promising direction, as we posit that stories can be instrumental in specific situations, rather than necessary in all circumstances.

DISCLOSURE STATEMENT

No potential conflict of interest was reported by the authors.

Table 3: GPT Results (continued from Table 2).

Model, Input, and Parameters	GPT response
Model: **BNE** Input: Stress Level: ['Very Low', 'Medium', 'Very High', 'Very High', 'Very High']	In the dimly lit confines of the tunnel, I find myself among a throng of pedestrians, a collective breath held in anticipation. My name is Michael, just an ordinary individual whose journey today has taken an unexpected turn. The air is thick with tension, a stark contrast to the mundane tranquility of my usual commute. At first, the situation seems manageable. My stress level is very low, a testament to the calm demeanor I've cultivated over the years. I assess the surroundings with a clear mind, noting the two exits that promise safety from this subterranean passage. The crowd around me is a sea of faces, each one etched with concern, but there's a shared sense of order as we begin to move. As we shuffle forward, whispers of urgency ripple through the crowd, and my stress level rises to medium. I can feel the shift in my heartbeat, a subtle reminder that the situation is evolving, becoming more precarious. I remind myself to breathe, to focus on the steady pace I need to maintain to reach one of the exits. Suddenly, the crowd's pace quickens, and a wave of panic crashes over us. My stress level skyrockets to very high. I'm swept up in the collective fear, my thoughts racing as I struggle to keep my footing. The exits seem impossibly distant, and the press of bodies is overwhelming. I can hear the sharp intake of breath from those around me, a chorus of anxiety that mirrors my own. The stress clings to me, a relentless force that refuses to ebb. Every step is a battle against the tide of my own emotions, which threaten to engulf me. I'm acutely aware of the danger, the need for escape pressing down on me with an almost physical weight. As I continue to navigate the chaos, my stress remains at a peak, a constant companion in this harrowing journey. The tunnel, once a simple passageway, has transformed into a labyrinth of fear.

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DATA AVAILABILITY STATEMENT

Data is openly accessible on a third-party repository at https://doi.org/10.5281/zenodo.10552705

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